

## Credit:

20% of total module mark

## Module Learning Outcomes Examined:

1. Understand, implement and use algorithms such as Viterbi decoding, and basic supervised classification.
2. Understand how NLE techniques can be used to design and implement applications such as text summarization, sentiment analysis and writing quality prediction.

## Submission Deadline:

Submission will be via the Faser online submission system. Please refer to Faser for the submission deadline. Please submit your `.ipynb`, readme file and a brief report as a pdf containing your code test output. You should submit these as a compressed zip file.

You should refer to the Undergraduate Students' Handbook for details of the departmental policy regarding late submission and university regulations regarding plagiarism; **the work handed in must be entirely your own**. It is expected that marking of the assignment will be completed within 3/4 working weeks of the submission deadline.

## Introduction:

Please write your code using Jupyter notebook and submit it as a ipynb file

Build a text classifier on the emotion classification dataset – a collection of English Twitter messages meticulously annotated with six fundamental emotions: anger, fear, joy, love, sadness, and surprise. You can use any classification method except the Naïve Bayes method and rule-based methods.

You should appropriately split your dataset into training and/or validation instances and a separate set of instances for testing your model. An example is to train your model on the first 90% of instances and testing your model on the last 10% of instances. Equally, you might randomise the instances and take test for an appropriate training / test split.

The Emotions dataset will be uploaded on the Moodle page for you to download however can also be obtained from the following Kaggle link:

Dataset Homepages: <https://www.kaggle.com/datasets/nelgiriyeewithana/emotions>

Some tutorials can be found here:

<https://www.kaggle.com/datasets/nelgiriyeewithana/emotions/code?datasetId=4403839&sortBy=voteCount>

Note, to access the tutorials you may need to create a free account on Kaggle.

## Assessment Criteria

Your code should include:

1. Read the file, incorporate the instances into the training set and testing set.
2. Pre-processing the text, you can choose whether you need stemming, removing stop words, removing non-alphabetical words. (Not all classification models need this step, it is OK if you think your model can perform better without this step, and you can give some justification in the report.)
3. Analysing the feature of the training data, report on the linguistic features.
4. Build a text classification model and train your model on the training set and test your model on the test set.

You can use the existing algorithms on Github or Kaggle (please include the references in your code).

You could implement algorithms based on CNNs, BERT (Bidirectional Encoder Representations from Transformers), Bidirectional Gated Recurrent Unit, LSTMs etc.

**However, you must not directly copy and paste their code!** You are not allowed to use the Naïve Bayes algorithm and VADER classifier, which you practiced in Lab 4).

5. Summarise the performance of your model using suitable classification evaluation metrics (You can gain additional marks if you have some graph visualisations).
6. You need to include a 'readme' file in your submission, in which you need to say:
  - What python version you used
  - libraries used in your assignment (also give the versions for these)
  - How to run your code
7. After you build your model and test it on the test set, you should write a report (no longer than three pages in A4, with Arial 11 fonts) to summarise your work.

Write your report like a mini-conference paper. You can learn how to format it from the example paper below:

Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. 2016. Hierarchical Attention Networks for Document Classification. In Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 1480–1489, San Diego, California. Association for Computational Linguistics

<https://aclanthology.org/N16-1174/>

## Additional Marks

Marks will be awarded:

- Have necessary comments on your code
- Have proper reference on your report
- Have graph visualization on your report
- Investigate more evaluation methods, for instance in addition to showing the Precision, Recall and F1 scores, also run the model multiple times and show the standard derivations for these scores. You can also apply other evaluation methods which you can find.

## WARNING AND ADVICE ABOUT POSSIBLE ACADEMIC OFFENCES

Your solutions should be your own unaided work. For more information, please see the University pages on plagiarism and the Academic Offences Procedures.

DO NOT COPY PROGRAM CODE FOR THIS ASSIGNMENT FROM ANOTHER STUDENT OR FROM THE INTERNET OR FROM ANY OTHER SOURCES. DO NOT LET OTHER STUDENTS COPY YOUR WORK

## Marks Scheme:

The marks for this assessment will be out of 100, broken down as follows:

### Working code, loading and preprocessing of data (20 pts)

---

- Compliant and working code
- Correctly loads text emotion classification dataset as directed
- Apply appropriate pre-processing on the dataset (stemming, removing stop words, removing non-alphabetical words) or provide clear justification as to why this is not needed based on the model used
- Report on the pre-processing steps applied on the dataset

### Feature analysis and reporting (20 pts)

---

- Analyse the features of the training set such as carrying out basic exploratory analysis on label distribution, length of texts (tokens), noise such as misspellings, lexical analysis such as word frequencies and their association to the emotional label or semantic analysis if embeddings are used
- Report on the features analysis carried out on the training dataset

### Text classification model (20 pts)

---

- Select and build a supervised text classification model of your choice
- Train the selected model on the training set and test your model on the test set
- Report on the details of the model you selected, any parameter configurations you applied, and the dataset split used with some justification

## Model evaluation (20 pts)

---

- Report on the performance of your model using suitable classification evaluation metrics such as Precision, Recall and F1

## Additional marks (20 pts)

---

- Have necessary comments on your code
- Have well formatted short report with proper references
- Have graph visualization of your model evaluation on your report
- Investigate more evaluation methods, for instance in addition to showing the Precision, Recall and F1 scores, also run the model multiple times and show the standard derivations for these scores. You can also apply other evaluation methods which you can find.

For each of the 5 marking criteria, marks will be awarded out of 20 points based on the following bands:

Poor:  $\leq 8$

Satisfactory: 9-10

Good: 11-13

Excellent: 14-16

Outstanding: 17-20